

1 Biomodelling.jl 2.0: mechanistic simulation of heritable gene-expression 2 states, cell division and drug response for single-cell benchmarking and 3 persister biology

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7 **Abstract**

8 Non-genetic heterogeneity, and in particular expression states that are inherited across cell divisions,
9 decides which cancer cells survive drug treatment, yet the simulators used to benchmark single-cell methods
10 model neither cell growth nor division, and therefore cannot generate such states from first principles. We
11 present Biomodelling.jl 2.0, a rewrite of a Julia framework for stochastic simulation of gene regulatory
12 networks inside growing, dividing and drug-treated cell populations. The framework couples exact, hybrid and
13 adaptive stochastic kinetics to exponential growth, volume-scaled transcription, gene replication, division with
14 binomial partitioning and promoter inheritance, and population control; adds dose schedules, state-dependent
15 death, growth inhibition and genetic perturbations; records every cell's lineage; converts molecule counts
16 into scRNA-seq, smFISH and time-lapse observations; and provides likelihood-free inference. We validate
17 the engine against exact stationary distributions and an independent implementation, show that growth and
18 division alone reproduce published results on cell-size scaling, partitioning noise, lineage-versus-population
19 statistics and cell-cycle-dependent bursting, and demonstrate that a single resistance gene with slow promoter
20 switching reproduces the signatures of drug-tolerant persisters reported in recent experiments and models:
21 dose-dependent decay with unchanged division times, correlated fates of related cells, preserved or collapsed
22 barcode diversity, loss of resistance when memory is disrupted during exposure, an intermediate-dose optimum
23 for drug-induced tolerance, and stable inheritance of a selected phenotype. Used as a ground-truth generator,
24 the framework evaluates memory-gene detection, network inference under the confounds of growth and
25 sequencing, imputation and perturbation-prediction baselines; used as an inference engine, it recovers kinetic
26 and drug parameters from snapshot and lineage data and quantifies the bias of ignoring division. Calibrated
27 to published time-lapse measurements of cisplatin-treated cells, the persister model predicts the fates of cells
28 at a held-out concentration, the biphasic kill curve, the extent of lineage correlations and the dose-invariance
29 of single-cell timing; used as a schedule testbed with a built-in optimiser, it reproduces the outcomes of
30 continuous versus intermittent BRAF/MEK inhibition in the SWOG S1320 trial and of intermittent dosing in
31 xenografts, attributes their difference to the fitness of resistant cells without drug, and evaluates standard
32 against dose-dense temozolomide regimens with pharmacokinetics. Biomodelling.jl 2.0 is open source (MIT) at
33 <https://github.com/ayoublasri/Biomodelling.jl>.

34 **Introduction**

35 Single-cell technologies have made non-genetic heterogeneity a central object of study in cell biology and drug
36 discovery. Isogenic cells differ in their transcriptional states, these states are often inherited across several divisions,

and the inherited states decide cell fate under stress: drug-tolerant persister cells that survive targeted or cytotoxic therapy arise from pre-existing or drug-induced expression states rather than from mutations (Shaffer et al. 2017, 2020; Harmange et al. 2023; Iyer et al. 2025; Lin et al. 2026). Lineage-resolved single-cell experiments now quantify this memory directly, showing that the fates of sisters and cousins are correlated, that clonal barcode diversity is preserved or collapsed depending on whether tolerance pre-exists or is induced, and that the timescale of expression fluctuations relative to the cell cycle determines which genes carry memory (Iyer et al. 2025; Lin et al. 2026; Ghosh et al. 2025). In parallel, mechanistic analyses of single-cell RNA-sequencing data have established that transcriptional kinetics can only be inferred correctly when gene copy number, cell age and partitioning at division are modelled explicitly (Sukys and Grima 2025; Zhang et al. 2025), and evaluations of “virtual cell” models for perturbation prediction have found that deep-learning approaches still fail to beat simple baselines, largely because they lack mechanism and causal structure (Ahlmann-Eltze et al. 2025; Vollenweider and Bühlmann 2026; Dibaeinia et al. 2026).

These developments create a demand that existing simulators do not meet. Simulators used to benchmark single-cell methods, such as SERGIO, dyngen, scMultiSim and GRouNdGAN (Dibaeinia and Sinha 2020; Cannoodt et al. 2021; Li et al. 2025; Zinati et al. 2024), model regulation and technical noise but not cell growth, volume-dependent transcription, gene replication or the partitioning of molecules at division; lineage simulators such as TedSim (Pan et al. 2022) generate division trees but attach phenomenological, largely Markovian expression to them. As a consequence none of them can produce expression memory, cell-size scaling, cell-cycle-dependent bursting or persister dynamics from first principles, and none provides the ground truth needed to test the methods that are now being developed for exactly those phenomena. Stochastic simulators of growing and dividing cells exist (Bertaux et al. 2018; Piho and Thomas 2026) but stop at the single-cell state: they do not provide drug response, lineage statistics, observation models, inference or benchmark generation.

Biomodelling.jl was introduced as a generator of synthetic scRNA-seq data from stochastic gene regulatory networks in growing and dividing cells, and was used to benchmark imputation methods for network inference (Lasri et al. 2022). Its distinguishing feature, the coupling of stochastic reaction kinetics to exponential growth, volume-scaled transcription and binomial partitioning at division, is precisely the mechanism behind the heritable states discussed above. Here we present version 2.0, a rewrite that turns this feature into a complete framework for the mechanistic study of heritable expression states, drug tolerance and single-cell benchmarking. The package adds (i) a typed model layer with mass-action, Hill and custom kinetics and explicit promoter states; (ii) exact, hybrid and adaptive stochastic kernels validated against analytic laws and an independent implementation; (iii) a population layer with sizer, adder and timer division control, binomial or over-dispersed partitioning, promoter inheritance, gene replication and constant-size, free-growth or logistic population control; (iv) a drug and perturbation layer with dose schedules, pharmacokinetics, state-dependent death, growth inhibition, drug-induced rate changes and genetic perturbations; (v) complete lineage records with heritability, autocorrelation, fluctuation-test and clonal-memory statistics; (vi) observation models for scRNA-seq, smFISH and time-lapse reporters with AnnData output; and (vii) likelihood-free inference and the exact telegraph likelihood.

We validate the engine against exact stationary distributions and against JumpProcesses.jl, show that growth, division and partitioning alone reproduce published results on cell-size scaling, partitioning noise, lineage-versus-population statistics and cell-cycle-dependent bursting, and demonstrate that a single resistance gene with slow promoter switching reproduces the quantitative signatures of drug-tolerant persisters reported in recent experimental and theoretical work: dose-dependent population decay with unchanged single-cell division times, correlated fates of related cells, preserved or collapsed barcode diversity, loss of resistance when memory is disrupted during exposure, an intermediate-dose optimum for drug-induced tolerance, and stable inheritance of a

selected phenotype. We then use the framework as a ground-truth generator to evaluate memory-gene detection, network inference under the confounds of growth and sequencing, imputation, and perturbation-prediction baselines, show that the same forward model supports parameter inference from snapshot and lineage data and quantifies the bias incurred by ignoring division, calibrate the persister model to published time-lapse data and validate it on held-out conditions, and finally use it, with a built-in schedule optimiser, to evaluate clinical dosing regimens and to search for better ones.

Results

A layered framework for stochastic simulation in growing, dividing and drug-treated populations

Biomodelling.jl 2.0 is organised in layers (Figure 1). A model is a list of reactions with mass-action, Hill or custom kinetics and named parameters; promoter states are explicit species grouped so that they are inherited at division and doubled at replication. Four kernels advance a cell’s state at fixed volume: the exact direct method, fixed-step tau-leaping, a hybrid scheme that falls back to the exact method whenever a leap would produce a negative count, and adaptive tau-leaping with critical-reaction handling. The population layer grows each cell’s volume exponentially, scales propensities with volume so that concentrations are invariant, replicates genes at a set point of the cycle, divides cells by a sizer, adder or timer rule with binomial (or over-dispersed) partitioning of molecules, and controls the population size (constant size by Moran-like replacement, free growth, or logistic). A drug and perturbation layer applies dose schedules, state-dependent death, growth inhibition, drug-induced rate changes and genetic perturbations. Every cell’s ancestry, fate and state at birth and division are recorded in a lineage table, from which heritability, autocorrelation, memory timescales, fluctuation-test and clonal-memory statistics are computed. Observation models convert true counts into scRNA-seq, smFISH or time-lapse measurements, and the same forward model drives likelihood-free inference. Table 1 compares these capabilities with those of the simulators most used to benchmark single-cell methods.

Table 1. Capabilities of single-cell simulators (✓ native; ○ partial or phenomenological; – absent). ABM.jl, AgentBasedModeling.jl; v1.x and v2.0, Biomodelling.jl versions.

Capability	SERGIO	dyngen	scMulti-Sim	GRouNd-GAN	TedSim	Lineage-Sim	ABM.jl	v1.x	v2.0
Mechanistic reaction kinetics (SSA / tau-leaping)	○	✓	○	–	–	–	✓	✓	✓
Gene regulatory networks with known ground truth	✓	✓	✓	✓	–	–	○	✓	✓
Cell growth, volume-scaled transcription	–	–	–	–	–	–	✓	✓	✓
Gene replication / cell- cycle copy number	–	–	○	–	–	–	–	–	✓
Division with molecular partitioning	–	–	–	–	○	○	✓	✓	✓
Heritable promoter states, expression memory	–	–	–	–	○	○	○	○	✓
Lineage tree with per-cell fates	–	–	–	–	✓	✓	○	–	✓
Drug response: dose schedules, state-dependent death	–	–	–	–	–	–	–	–	✓
Genetic perturbations (knockdown / overexpression)	✓	✓	○	✓	–	–	○	–	✓

Capability	SERGIO	dyngen	scMulti-Sim	GRouNd-GAN	TedSim	Lineage-Sim	ABM.jl	v1.x	v2.0
scRNA-seq observation model	✓	✓	✓	✓	✓	✓	–	○	✓
smFISH / time-lapse observation	–	–	–	–	–	–	–	–	✓
Parameter inference from data	–	–	–	–	–	–	–	–	✓
AnnData output	✓	✓	✓	✓	○	✓	–	–	✓

The engine is exact and agrees with an independent implementation

We first verified the kernels against exact stationary laws (Figure 2). For a birth-death process the simulated distribution of 50 000 cells is indistinguishable from the Poisson law (Figure 2a); for the telegraph model in an intermediate and in a bursty regime it matches the Beta-Poisson distribution of Peccoud and Ycart (1995) (Figure 2b); and for bursty protein production it matches the negative binomial (Figure 2c). Across kernels, Kolmogorov-Smirnov distances to the exact law with 20 000 cells are below the 99% critical value for the direct method and for the hybrid and adaptive schemes at their default tolerances, at wall-clock costs of 0.25 s to 0.91 s per 10^4 cells (Figure 2d). Simulating the same telegraph model with JumpProcesses.jl gives distributions whose two-sample Kolmogorov-Smirnov statistic against Biomodelling.jl is 0.0034 (99% critical value 0.0163; Figure 2e), with a 6.2-fold (in favour of JumpProcesses.jl) difference in serial wall-clock time. Population simulations scale linearly with the number of cells and approximately linearly with the number of reactions: 2.6 s for 5000 cells $\times$ 10 telegraph genes $\times$ 200 steps (hybrid kernel) with four threads (Figure 2f).

Growth, division and partitioning generate expression memory, size scaling and cell-cycle effects

The physiology of the population layer is sufficient to produce the properties of single-cell data that phenomenological simulators must impose by hand (Figure 3). In a single lineage of a slowly switching gene, mRNA and protein track volume within a cycle and are halved at division, while the promoter state, inherited by both daughters, persists over many divisions (Figure 3a). For a fast-switching gene without replication, transcript numbers scale linearly with cell volume so that concentration is homeostatic; adding mid-cycle gene replication doubles transcription in the second half of the cycle and lifts counts of large cells above the volume-scaling line (Figure 3b), the copy-number effect that mechanistic fits to scRNA-seq data must account for (Sukys and Grima 2025). Correlations of protein concentration at division between mothers and daughters, sisters and cousins decrease from 0.94 to 0.02 as the promoter switching rate crosses the inverse cell-cycle time (Figure 3c), with sisters more correlated than mother-daughter pairs and cousins less, and mRNA less heritable than protein. The memory timescale extracted from the generation-resolved lineage autocorrelation is set by the promoter switching time and is nearly independent of the cell-cycle time as long as memory outlasts the cycle; when the two are comparable the estimate is bounded below by one cell cycle, the resolution of the autocorrelation (Figure 3d). Following Zhang et al. (2025), the coefficient of variation of a stable protein’s concentration is larger across the population than along single lineages, whether or not cell-cycle times are noisy, and over-dispersed partitioning adds variability in both perspectives (Figure 3e). Finally, telegraph parameters fitted separately to cells before and after replication differ by the expected factor of two in burst frequency, and a fit that ignores the cell cycle returns intermediate values (Figure 3f).

A single resistance gene with slow promoter switching reproduces persister biology

To test whether heritable expression states suffice to reproduce the observed phenomenology of drug-tolerant persisters, we coupled a resistance gene with slow promoter switching (memory timescale 45 time units, cell-cycle time 20) to a death hazard that acts on the drug dose and is suppressed by the concentration of the gene's protein product (Methods). Under continuous, pulsed and reduced-dose schedules the population first collapses as low-expressing cells die and then regrows from the high-expressing survivors (Figure 4a); kill curves are biphasic with a surviving fraction that decreases with dose (Figure 4b). The net rate of population decline over the first 30 time units of treatment rises from 0.023 to 0.050 per time unit between doses 0.5 and 2 (2.1-fold), while the mean division time of cells born under drug changes by only 0.0% and the mean time to death of killed cells only moves from 8.5 to 6.2 time units (Figure 4c), reproducing the observation of Iyer et al. (2025) that population decay depends strongly on dose even though single-cell division and death times barely do, because fate is decided by cell states present before the drug. Accordingly, related cells born in the 25 time units before the drug share the fate of their lineages (extinction, or survival of at least one descendant): a cell whose sister founded a surviving lineage does so itself with probability 0.39, against a marginal probability of 0.08 (4.8-fold enrichment over 1830 sister pairs; 2.9-fold for cousins), whereas in the fast-switching control, in which survival is rarer, the conditional and marginal probabilities coincide (0.03 versus 0.02; Figure 4d). When tolerance pre-exists, survivors descend from a minority of founder clones and the effective number of clones drops from 491 to 89, whereas when tolerance is drug-induced (the drug raises the promoter activation rate of every cell) barcode diversity is largely preserved (165 of 492; Figure 4e), the discriminating signature used in barcoding experiments (Shaffer et al. 2017; Iyer et al. 2025). Disrupting expression memory changes this outcome: accelerating promoter switching 20-fold from two cell-cycle times before the drug, as memory-disrupting epigenetic compounds do (Harmange et al. 2023), leaves the number of surviving clones unchanged when the acceleration stops at the moment the drug arrives (98 versus 120 clones), because it reshuffles which cells are in the resistant state without changing their number, but reduces it to 7 clones (10 cells, against 792) when the acceleration continues during exposure (Figure 4f), because resistant cells now leave the protected state within a fraction of the time they need to establish a colony: in the model, memory must be disrupted while the drug is present. A transient drug pulse enriches high-expressing cells and, with slow switching, the enrichment persists for many generations after the drug is removed, whereas with fast switching it relaxes within a few cycles (Figure 4g): the phenotypic selection with stable inheritance of the original MGMT model (Lasri et al. 2020). Finally, scanning release periods and doses of intermittent schedules (Figure 4h; net growth rate of the population over a 200-time-unit course), the most effective schedule is release period 0, dose 2 (growth rate +0.004 per time unit) for pre-existing tolerance, release period 10, dose 2 (growth rate -0.016 per time unit) when the resistant state carries a fitness cost, and release period 10, dose 1 (growth rate +0.009 per time unit) for drug-induced tolerance (the drug raising the promoter activation rate 100-fold at dose 1). Without a fitness cost, therefore, nothing beats continuous dosing at the highest dose and every release period lets the resistant subpopulation regrow. With a fitness cost of the resistant state the population declines under every schedule with dose ≥ 0.5 , and release periods of up to 10 time units are as effective as continuous dosing (-0.013 per time unit under continuous dose 2), because cells that leave the resistant state during the holiday are killed on re-exposure. For drug-induced tolerance the ranking of doses reverses: the highest dose induces tolerance in more cells than it kills, so dose 1 outperforms dose 2 at every release period (release period 10, dose 1 (growth rate +0.009 per time unit), against +0.012 for continuous dose 2), the intermediate-dose optimum reported by Corigliano et al. (2025); the benefit of release periods reported in that work does not appear in this model, in which the induced state decays at the same rate with and without drug, so that a holiday only postpones the death of reverting cells while all cells grow.

Ground truth for memory genes, network inference, imputation and perturbation prediction

Because every cell's clone, lineage and true state are known, the framework supplies ground truth that observational data cannot (Figure 5). For 40 independent genes whose memory times span three decades, MemorySeq-style clonal variance scores computed on true concentrations rise from 1.0 for genes with memory times far below the cell-cycle time to 29.6 for the slowest genes, and the ranking is preserved when scores are computed on sequenced counts at 20% capture efficiency (Figure 5a); the score separates genes with memory times above the cycle time from the rest with an area under the receiver operating curve of 1.00 (true) and 0.99 (sequenced). For a 30-gene network simulated at fixed volume, in a dividing population and after sequencing, correlation-based and GENIE3 inference lose most of their accuracy in the dividing population when raw counts are used (area under the precision-recall curve 0.52 to 0.14 for Pearson correlation and 0.32 to 0.13 for GENIE3, against 0.08 for a random ranking), because volume and gene copy number induce positive correlations between all genes. Normalising counts to concentrations removes the volume effect but recovers little (0.18 and 0.18), since the dependence of expression on copy number and cell age remains, whereas regressing expression on the true cell-cycle covariates (copy number and age, which the simulation provides and which cell-cycle scoring approximates in real data) before inference recovers most of the loss (0.46 and 0.30). Sequencing at 15% capture efficiency lowers accuracy further (0.11 for raw and 0.24 for library-size-normalised counts with Pearson correlation; 0.07 and 0.08 with GENIE3; Figure 5b). Imputation with kNN-smoothing and MAGIC before inference changes the area under the precision-recall curve by +0.05 to +0.21 (Pearson correlation) (Figure 5c), consistent with the conclusions of the original benchmark (Lasri et al. 2022). Finally, knocking down the five most connected regulators of a 20-gene network produces true fold changes that a no-change baseline and the propagation of observational correlations predict with coefficients of determination of -0.16 and -0.23 respectively (Figure 5d), illustrating how causal, out-of-distribution ground truth can be generated for the evaluation of perturbation-prediction models (Ahlmann-Eltze et al. 2025; Vollenweider and Bühlmann 2026).

The forward model supports inference and quantifies the bias of ignoring division

On a snapshot of 2 000 non-dividing cells, ABC-SMC with log-uniform priors over three decades and the Wasserstein distance between simulated and observed counts (Bernton et al. 2019) concentrates on the true telegraph parameters and agrees with the exact maximum-likelihood estimate (Figure 6a; posterior medians (0.31, 0.60, 20.9) for truth (0.3, 0.6, 20)), with the broadest posterior on k_{off} , the parameter that a single snapshot constrains least. On a snapshot of a growing and dividing population with volume-scaled transcription and mid-cycle replication, the naive telegraph fit is biased ($k_{\text{on}} = 0.47$, $k_{\text{off}} = 1.95$, $k_{\text{tx}} = 71.5$) because volume and copy number inflate counts and their dispersion, whereas ABC with the population simulator as forward model recovers the truth ((0.30, 0.68, 20.8); Figure 6b). Drug parameters can be inferred from population data in the same way: from a kill curve and the concordance of sister fates, the posterior over the maximal death rate and the protection threshold is a ridge (weighted correlation of $\log h_{\text{max}}$ and $\log K$ of -0.85) on which the truth (0.5, 150) lies (Figure 6c), and posterior predictive kill curves simulated at the posterior median reproduce the observed one (Figure 6d): the two summaries constrain the product of death rate and unprotected fraction, and separating the two parameters requires a measurement of the protective protein, an example of the experimental-design questions the forward model can answer before an experiment is run.

The persister model calibrates to laboratory time-lapse data and predicts held-out conditions

A simulator is only useful for treatment questions if it can be fitted to the data a laboratory actually produces and then predicts what it was not fitted to. We therefore calibrated the resistance-gene model to the time-lapse measurements of Iyer et al. (2025), who tracked U2OS cells for three days under 7, 10 and 13 μM cisplatin and recorded, for every cell present at drug addition, whether it died, divided or survived without dividing (Table S3). Six parameters (promoter switching rates, maximal death rate, EC_{50} and Hill coefficient of death, IC_{50} of growth arrest) were fitted to the fate fractions at 7 and 13 μM by simulated moments with common random numbers, with a weak prior keeping the memory of the resistant state near the two to three generations that the same study inferred from lineage correlations (Methods). The calibrated model (Figure 7f; memory 2.8 generations, resistant fraction 15.2%, EC_{50} 16.7 μM) reproduces the training fractions with a root-mean-square error of 0.071 and predicts the held-out 10 μM condition with an error of 0.093 (observed died, divided and survived fractions 0.28, 0.31 and 0.41; predicted 0.39, 0.20 and 0.41; Figure 7a). Properties that were not fitted can then be compared with the data. Death fates are correlated between sisters (0.09), first cousins (0.06) and second cousins (0.05) and the correlation decays to 0.02 for third cousins and 0.00 for unrelated cells (Figure 7c), the pattern reported for HCT116 lineages. The mean time to death changes by only 8% between 7 and 13 μM while the death fraction rises from 0.13 to 0.59 (Figure 7d), the dose-invariance of single-cell timing that motivated the original study, and arises here because the dose changes the fraction of unprotected cells rather than their kinetics. The predicted kill curves (Figure 7b) are steeper at 13 μM in their first two days than in their second half (1.5-fold), but the calibrated memory of 2.8 generations is too short to produce the plateau that HCT116 cells reach after about 100 h: in the model, cells leave the resistant state within days and are then killed. Reproducing the plateau requires a longer memory or a drug-induced stabilisation of the resistant state, which is exactly the kind of conclusion a calibrated forward model can draw and a lineage-tracing experiment can test. The calibration landscape (Figure 7e) shows how the fitted death rate and EC_{50} are constrained by the fate fractions alone; the same protocol, with the same functions, applies to any observable that a laboratory records.

Dose and schedule optimisation against clinical regimens

With a calibrated model the framework becomes a schedule testbed. We first asked which dosing schedule minimises regrowth for a melanoma-like population with a rare, heritable pre-resistant state (Shaffer et al. 2017) treated at a fractional-killing dose, comparing continuous dosing with the intermittent schedule of the SWOG S1320 trial, in which patients received a continuous 8-week lead-in and were then randomised to continuous dosing or to cycles of 3 weeks off and 5 weeks on (Algazi et al. 2020), and with adaptive therapy, in which the drug is withdrawn when the population has shrunk to half its initial size and resumed when it regrows to the initial size (Zhang et al. 2017). When the resistant state carries no fitness cost, as the lineage data of Iyer et al. (2025) indicate for pre-existing tolerance, continuous dosing gives the longest time to progression ([ttp_cont_nocost] weeks against [ttp_int_nocost] for the intermittent schedule and [ttp_adapt_nocost] for adaptive therapy; Figure 8a,c), the ranking observed in the trial, where continuous dosing gave a median progression-free survival of 9.0 months against 5.5 months for intermittent dosing. When resistant cells pay a fitness cost in the absence of drug, the drug addiction that made intermittent dosing superior in patient-derived xenografts (Das Thakur and Stuart 2013), the ranking reverses: the intermittent schedule and adaptive therapy delay progression to [ttp_int_cost] and [ttp_adapt_cost] weeks against [ttp_cont_cost] for continuous dosing, with adaptive therapy using [dose_adapt_cost] of the cumulative dose (Figure 8b,c). The optimiser makes the dependence explicit: over intermittent schedules of any period and duty cycle, the best schedule without a fitness cost is [best_nocost], and

with a fitness cost release period 10, dose 2 (growth rate -0.016 per time unit) (Figure 8d). The model therefore reproduces both clinical and preclinical outcomes and attributes their difference to a measurable property of the resistant cells, their fitness without drug, which the framework can estimate from growth data before a schedule is chosen.

The second case study concerns glioblastoma, where the RTOG 0525 trial compared the standard temozolomide regimen (150-200 mg/m² on days 1-5 of 28-day cycles) with a dose-dense regimen (75-100 mg/m² on days 1-21) that delivers about twice the cumulative dose, and found no difference in overall survival (16.6 against 14.9 months) for either MGMT-methylated or unmethylated tumours (Gilbert et al. 2013). Simulating both regimens with one-compartment pharmacokinetics (elimination half-life 2.1 h; Ostermann et al. (2004)) on populations in which 1% (methylated) or 30% (unmethylated) of cells express MGMT, with MGMT protein protecting cells from the drug (the phenotypic-selection model of Lasri et al. (2020)), [gbm_summary] (Figure 8e,f). At equal cumulative dose the number of dosing days per cycle [gbm_days_summary] (Figure 8g). These simulations are not fits to the trials, whose end points depend on much that the model omits, but they show how the framework turns a proposed regimen into a quantitative, mechanism-resolved prediction, and how it identifies which biological measurement, the fitness of resistant cells without drug or the fraction of cells protected, decides between regimens.

Discussion

Biomodelling.jl 2.0 occupies a niche that the single-cell field currently lacks: a mechanistic simulator in which expression memory, cell-size effects, cell-cycle dependence and drug-tolerant persistence are consequences of explicit biophysics rather than assumptions of a generative model. Three uses follow. First, as a hypothesis engine for non-genetic drug tolerance, it lets one ask which schedule, which combination or which memory-modulating pretreatment minimises the long-term fitness of a treated population, with cell-level rules that can be matched to time-lapse and lineage-tracing data. Second, as a ground-truth generator, it produces data with known memory genes, known networks, known perturbation effects and known lineage trees, which is what methods such as MemorySeq-style scores, Power-Seek, lineage-aware models and virtual-cell benchmarks require to be evaluated fairly, and which existing simulators cannot provide. Third, as a likelihood-free inference engine, it allows mechanistic parameters to be estimated from the kinds of data that are now routinely collected, while quantifying the bias of simpler models.

The calibration and schedule studies illustrate the workflow we propose for in silico single-cell pharmacology: declare the training and validation observables before fitting, fit by simulated moments or ABC with common random numbers, validate on conditions and statistics that were not fitted, and only then compare regimens, reporting outcomes that map onto clinical end points (time to progression from the nadir, depth of response, cumulative dose). The two case studies show why mechanism matters: the same schedule question, continuous or intermittent dosing, has opposite answers depending on whether resistant cells pay a fitness cost without drug, and the trial and xenograft outcomes that seemed to contradict each other are both reproduced once that property is set. Because the framework simulates the lineage-resolved, single-cell data from which such properties are measured, it can also say which experiment would resolve the question before a trial is designed.

Several limitations delimit the current release. Cells are well-mixed compartments without spatial structure or cell-cell signalling; the pharmacology is a one-compartment model; growth is exponential with phenomenological size control; and the inference layer, although general, relies on user-chosen summary statistics, so posterior calibration should be checked with predictive simulations as in Figure 6. The persister case study is deliberately minimal, a single resistance gene with slow switching; its calibration used fate fractions transcribed from a published study

rather than raw tracks, and the clinical case studies use literature values for cell-cycle times, resistant fractions and pharmacokinetics rather than patient-specific measurements, so their conclusions are about mechanisms and rankings, not about absolute survival times. The Catalyst.jl bridge planned for interoperability with the SciML ecosystem is not part of this release. Finally, the benchmark results should be read as demonstrations of what the framework makes possible rather than as verdicts on the methods compared, which were run with default settings on small networks.

Future work will add spatial coupling through neighbourhood-dependent rates, cell-cell communication, chromatin states with their own inheritance rules, and neural simulation-based inference on top of the existing forward model. Because the framework is deterministic given a seed, fast enough to simulate tens of thousands of cells in minutes, and outputs AnnData files, it can be dropped into existing single-cell analysis pipelines as a source of mechanistically grounded synthetic data, and into virtual-cell training and evaluation as a source of causal, out-of-distribution contexts.

Methods

Reaction model and propensities

A model is a set of species and reaction channels. Species are molecules (partitioned at division) or promoter states (inherited). For a reaction with reactant counts x_i , stoichiometric coefficients ν_i , rate constant k and volume exponent e , the mass-action propensity is

$$a = k V^e \prod_i \binom{x_i}{\nu_i},$$

where by default $e = 1 - \sum_i \nu_i$, so that zero-order production scales with volume, first-order reactions are volume independent and bimolecular reactions scale as $1/V$; this keeps concentrations invariant under growth. Regulated reactions multiply this by $b + (1 - b)F(\mathbf{c})$, where b is a basal fraction and F combines Hill terms of the regulator concentrations $c_j = x_j/V$, $h_j = c_j^{n_j}/(K_j^{n_j} + c_j^{n_j})$ for activators and $1 - h_j$ for inhibitors, either as a product (AND logic) or as $1 - \prod_j (1 - h_j)$ (OR logic). Promoter switching is represented by explicit promoter species (for instance $G_{\text{off}} \rightleftharpoons G_{\text{on}}$) grouped into promoter groups. Custom propensities can be supplied as functions of counts, volume, parameters and time.

Stochastic simulation kernels

Within one update step the cell volume and gene copy number are held fixed and the reaction network is advanced by one of four kernels: the direct method (Gillespie 1977) with propensities updated through a species-reaction dependency graph; fixed-step Poisson tau-leaping (Gillespie 2001); a hybrid scheme in which a leap that would drive any species negative is rejected and the interval is instead simulated exactly (this is the exact version of the “switching” scheme of version 1); and adaptive tau-leaping with the step-size selection of Cao et al. (2006) and the critical-reaction partition of Cao et al. (2005), falling back to a fixed number of exact events when the proposed leap is shorter than $10/a_0$. Every kernel takes an explicit random number generator.

Cells, growth, replication and division

Each cell carries its volume V , age, generation, lineage identifiers, gene copy state, growth rate, molecule counts, a cell-specific parameter vector and its own random number generator (seeded from the parent’s generator, which

337 makes results independent of the number of threads). Volume grows exponentially, $\dot{V} = \lambda V$, with an optional
338 log-normal spread of λ between cells. Division is triggered by a sizer ($V \geq V_{\text{div}}$), an adder ($V - V_{\text{birth}} \geq \Delta$) or
339 a timer ($\text{age} \geq T$), each with a per-cell noise factor drawn at birth. At division the first daughter inherits a
340 volume fraction $f \sim \mathcal{N}(0.5, \sigma^2)$ (clipped to $[0.05, 0.95]$) and each molecule is assigned to it independently with
341 probability f (binomial partitioning); a beta-binomial option models over-dispersed partitioning. Promoter groups
342 are inherited by both daughters when unreplicated and split one copy per daughter, in random assignment, when
343 replicated. Gene replication, if enabled, doubles the promoter counts (and the propensity of reactions flagged as
344 copy-number dependent) once the cell has completed a set fraction of its cycle, and is reset at division. Populations
345 can be started from per-founder states and volumes (for instance a snapshot of a burnt-in constant-size population).
346 Population control is either constant size (every second daughter replaces a uniformly chosen cell, the behaviour
347 of version 1), free growth (all daughters kept, with uniform subsampling above a cap and book-keeping of the true
348 size) or logistic (free growth with an additional hazard $\lambda N/K$).

349 Drug and perturbation layer

350 A perturbation combines a dose schedule $d(t)$ (constant, pulsed, piecewise, one-compartment bolus pharmacoki-
351 netics or an arbitrary function) with effects evaluated for every cell at every step: a death hazard

$$h = h_{\max} \frac{d^m}{\text{EC}_{50}^m + d^m} \cdot \frac{K^q}{K^q + c_P^q},$$

352 where c_P is the concentration of a protective species (survival probability over a step is $e^{-h \Delta t}$); growth inhibition
353 by $1/(1 + (d/\text{IC}_{50})^m)$; a dose-independent fitness cost of a cell state, $1 - c_{\max} c_P^q/(K^q + c_P^q)$, on the growth rate;
354 multiplicative modulation of any parameter by a function of the dose; and genetic perturbations (knockdown,
355 knockout, over-expression) of any parameter in a random fraction of cells, the flag being inherited by daughters.

356 Lineage statistics

357 The lineage table records, for every cell, its parent, founder (clone), generation, birth and end times, fate (divided,
358 died, removed, alive) and its volume and state at birth and at the end of its life. From it we compute Pearson
359 correlations of expression at division between mothers and daughters, sisters and cousins; the lineage autocorrelation
360 r_g between a cell and its ancestor g generations earlier, and the memory timescale τ from $r_g = e^{-g/\tau}$; single-lineage
361 traces by following the ancestral line of a surviving cell; the coefficient of variation squared of concentration across
362 the population versus along single lineages; Luria-Delbrück fluctuation tests (variance of a phenotype across clones
363 grown from single cells compared with the binomial expectation); and MemorySeq-style clonal scores (Shaffer et
364 al. 2020), the variance of clone-mean expression divided by its permutation-null expectation.

365 Observation models

366 True counts are converted to observations by a scRNA-seq model in which each cell captures every molecule
367 independently with a Beta-distributed efficiency (mean and coefficient of variation set by the protocol), optional
368 multinomial down-sampling to a Poisson-distributed depth, optional additional dropout and batch scaling; an
369 smFISH model with a fixed detection efficiency; and a time-lapse model that samples a species along lineages at a
370 fixed interval with additive or multiplicative measurement noise. Because counts scale with volume, library size
371 correlates with cell size, reproducing the transcriptome-size confound of real data. Outputs are written as CSV or
372 as AnnData-compatible HDF5 files.

Inference

Snapshot counts of a gene in non-dividing cells follow the Beta-Poisson law of the telegraph model (Peccoud and Ycart 1995), which we evaluate through Kummer’s confluent hypergeometric function with rescaled series summation, and maximise by a grid-initialised Nelder-Mead search in log-parameter space. For any model, including populations, we implement sequential Monte Carlo approximate Bayesian computation (Toni et al. 2009) with an adaptive tolerance schedule (the α quantile of the previous generation’s distances), Gaussian perturbation kernels of variance twice the weighted population variance, importance weights against the prior, and parallel evaluation of the user-supplied distance function; for count distributions we use the Wasserstein-1 distance between simulated and observed samples (Bernton et al. 2019).

Calibration to laboratory data and validation strategy

Reference values were transcribed from the text and tables of the cited articles into `paper/data/` with their provenance (Table S3). For the cisplatin case study the model is a resistance gene with slow promoter switching whose protein product protects against a dose-dependent death hazard and against a dose-dependent growth arrest (Methods above), with a 24-h cell cycle. Its six free parameters (activation and inactivation rates, maximal death rate, EC_{50} and Hill coefficient of death, IC_{50} of growth arrest) were fitted by the method of simulated moments: the root-mean-square difference between simulated and observed fractions of cells that died, divided or survived without dividing during 72 h of drug, for the training concentrations (7 and 13 μ M), plus a weak penalty keeping the memory of the resistant state near the 2-3 generations inferred from lineage correlations, was minimised with the bounded optimiser described below using common random numbers (the same seeds for every candidate). Validation used only quantities not fitted: the fate fractions at the held-out concentration (10 μ M), the biphasic kill curve and its plateau, the extent of lineage correlations in fate (sisters, first, second and third cousins) and the invariance of single-cell division and death times across concentrations. The protocol is deliberately generic: any observable computable from a `PopulationResult` (kill curves, fate fractions, lineage correlations, snapshot distributions) can enter the distance, and the split between training and validation conditions is declared before fitting.

Clinical regimens, treatment outcomes and schedule optimisation

Regimens are built from the schedule primitives of the perturbation layer: `PulsedDose` with a finite number of cycles, `PiecewiseDose` for arbitrary calendars (the S1320 schedule), `daily_boluses` and `cycle_days` for daily oral doses with one-compartment pharmacokinetics, and `AdaptiveDose`, a feedback schedule that withdraws the drug when the population falls below a fraction of its initial size and resumes it when the population regrows above another fraction, evaluated with the current population size at every step (Zhang et al. 2017). Outcomes computed from a simulation are the net growth rate, the depth of response ($\log_{10}$ of the ratio of the initial to the smallest population), the time to progression (first time the population exceeds a threshold, 1.2, times its running minimum, the analogue of RECIST progression from the nadir; censored at the end of the simulation), the extinction probability over replicates and the cumulative dose actually applied. `optimize_schedule` minimises any such outcome over a box of schedule parameters: a log-spaced grid (or a Latin hypercube when the grid is large) is evaluated with the same random seeds for every candidate, and the best point is refined by Nelder-Mead in a bounded transform of the box; constraints such as a fixed cumulative dose enter as penalties or as the parameterisation (for instance a per-day dose equal to the budget divided by the number of dosing days). The melanoma case study uses a 30-h cell cycle, a pre-resistant fraction of 1:200 with a memory of five generations (Shaffer et al. 2017, 2020), drug effects at the fractional-killing dose (state-dependent death and growth arrest,

and a ten-fold stabilisation of the resistant state by the drug, the drug-induced reprogramming of Shaffer et al. (2017)), with or without a 50% fitness cost of the resistant state (Das Thakur and Stuart 2013); the temozolomide case study uses a 50-h cell cycle (Lasri et al. 2020), an elimination half-life of 2.1 h (Ostermann et al. 2004), the regimens of RTOG 0525 (Gilbert et al. 2013) (200 mg/m² on days 1-5 or 100 mg/m² on days 1-21 of 28-day cycles, six cycles) and MGMT-expressing fractions of 1% and 30% for methylated and unmethylated promoters.

Random regulatory networks

Random networks are built with a chosen number of activating and inhibiting edges under Erdős-Rényi or preferential-attachment (scale-free) sampling; regulation acts on the transcription rate of constitutive genes or on the promoter activation rate of telegraph genes through AND-combined Hill functions; per-gene rate constants can be drawn log-uniformly from ranges.

Validation protocols

Exactness was tested by Kolmogorov-Smirnov distances between simulated stationary distributions and the analytic laws (Poisson for birth-death, Beta-Poisson for the telegraph model, negative binomial for bursty protein production (Friedman et al. 2006; Shahrezaei and Swain 2008)), by agreement between kernels, and by a two-sample comparison against JumpProcesses.jl (Rackauckas and Nie 2017) on the same model. Population-level properties were compared with the analytic expectations for cell-size scaling and concentration homeostasis (Bertaux et al. 2018), partitioning noise (Huh and Paulsson 2011), lineage versus population statistics (Zhang et al. 2025) and cell-cycle-dependent burst parameters (Sukys and Grima 2025). Persister simulations were compared with the quantitative signatures reported by Iyer et al. (2025), Corigliano et al. (2025) and Harmange et al. (2023), and with the phenotypic-selection model of Lasri et al. (2020). All scripts, seeds and outputs are in the `paper/` directory of the repository.

Software

Biomodelling.jl 2.0 is written in Julia (≥ 1.10) and depends on Distributions.jl, PoissonRandom.jl, QuadGK.jl, Optim.jl, SpecialFunctions.jl and StatsBase.jl; HDF5.jl is an optional extension. Benchmarks used absolute Pearson and Spearman correlations and scikit-learn random forests re-implementing GENIE3 (Huynh-Thu et al. 2010), MAGIC (Dijk et al. 2018) and kNN-smoothing (Wagner et al. 2018). The test suite (Aqua quality checks and statistical tests) runs under GitHub Actions on Julia 1.10 and the latest release.

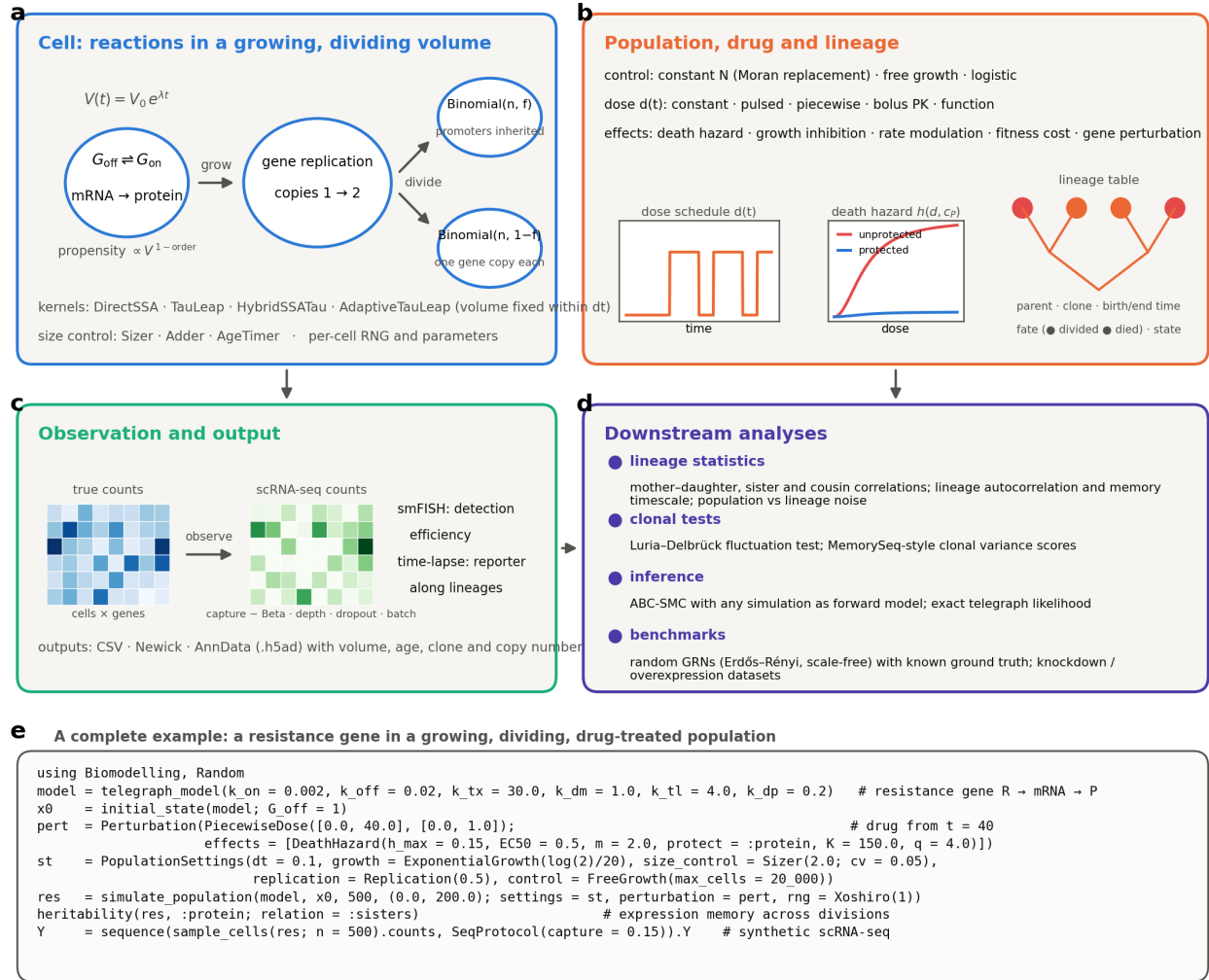


Figure 1. The Biomodelling.jl 2.0 framework. (a) Each cell runs the stochastic kinetics of a reaction network (mass-action, Hill or custom propensities, explicit promoter states) at its current volume; volume grows exponentially, genes replicate at a set point of the cycle, and division follows a sizer, adder or timer rule with binomial partitioning of molecules and inheritance of promoter states. (b) Population control (constant size, free growth or logistic), the drug and perturbation layer (dose schedules, state-dependent death, growth inhibition, rate modulation, genetic perturbations) and the lineage table that records every cell's ancestry, fate and state at birth and division. (c) Observation models (scRNA-seq with capture, depth, dropout and batch effects; smFISH; time-lapse reporters) and outputs (CSV, Newick, AnnData). (d) Downstream uses: lineage statistics, fluctuation tests, clonal memory scores, likelihood-free inference and benchmark generation. (e) A complete example in ten lines of code.

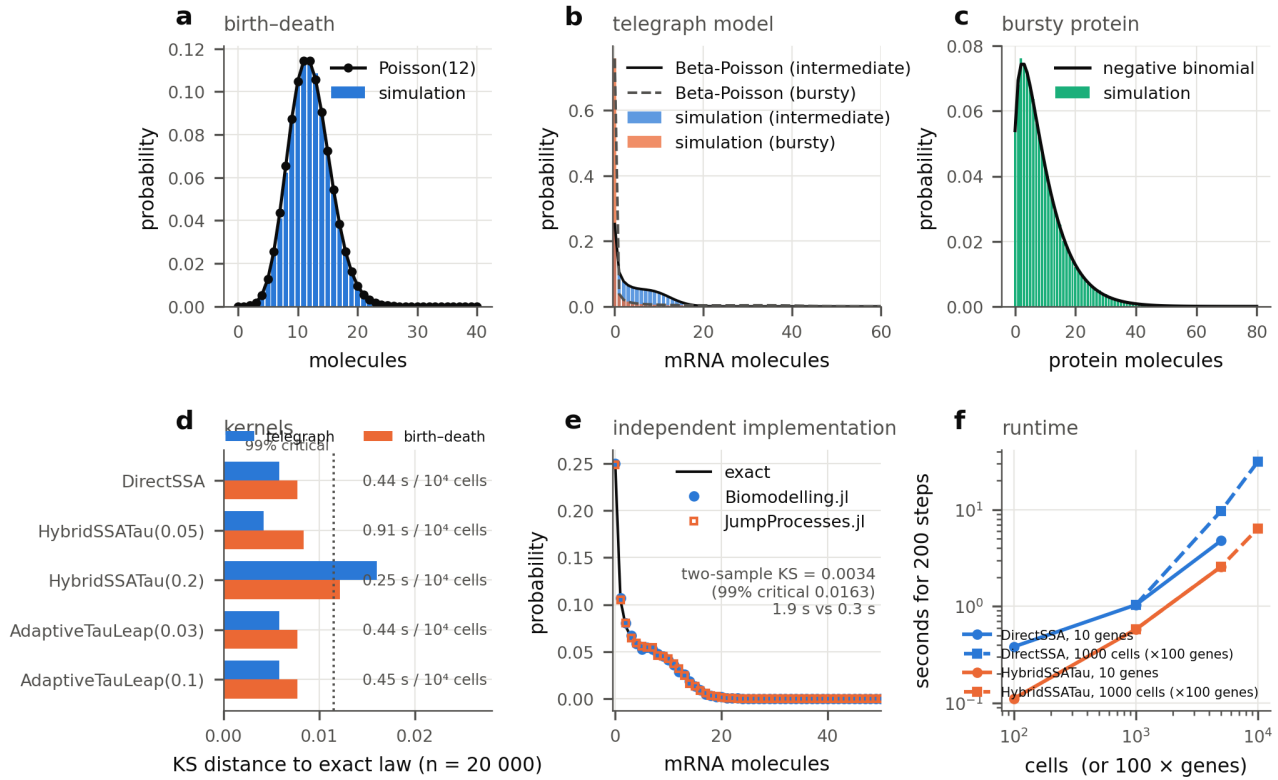
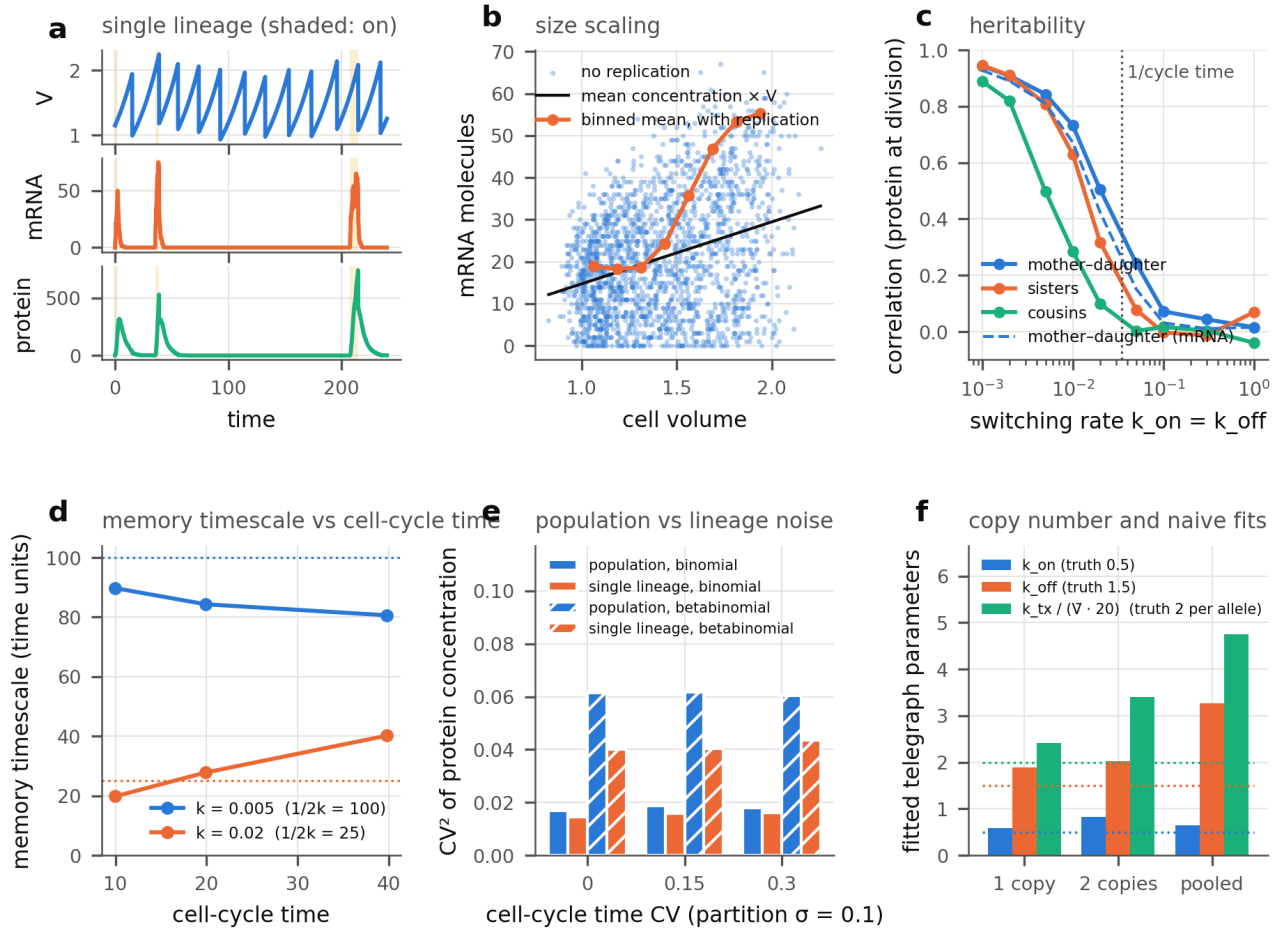
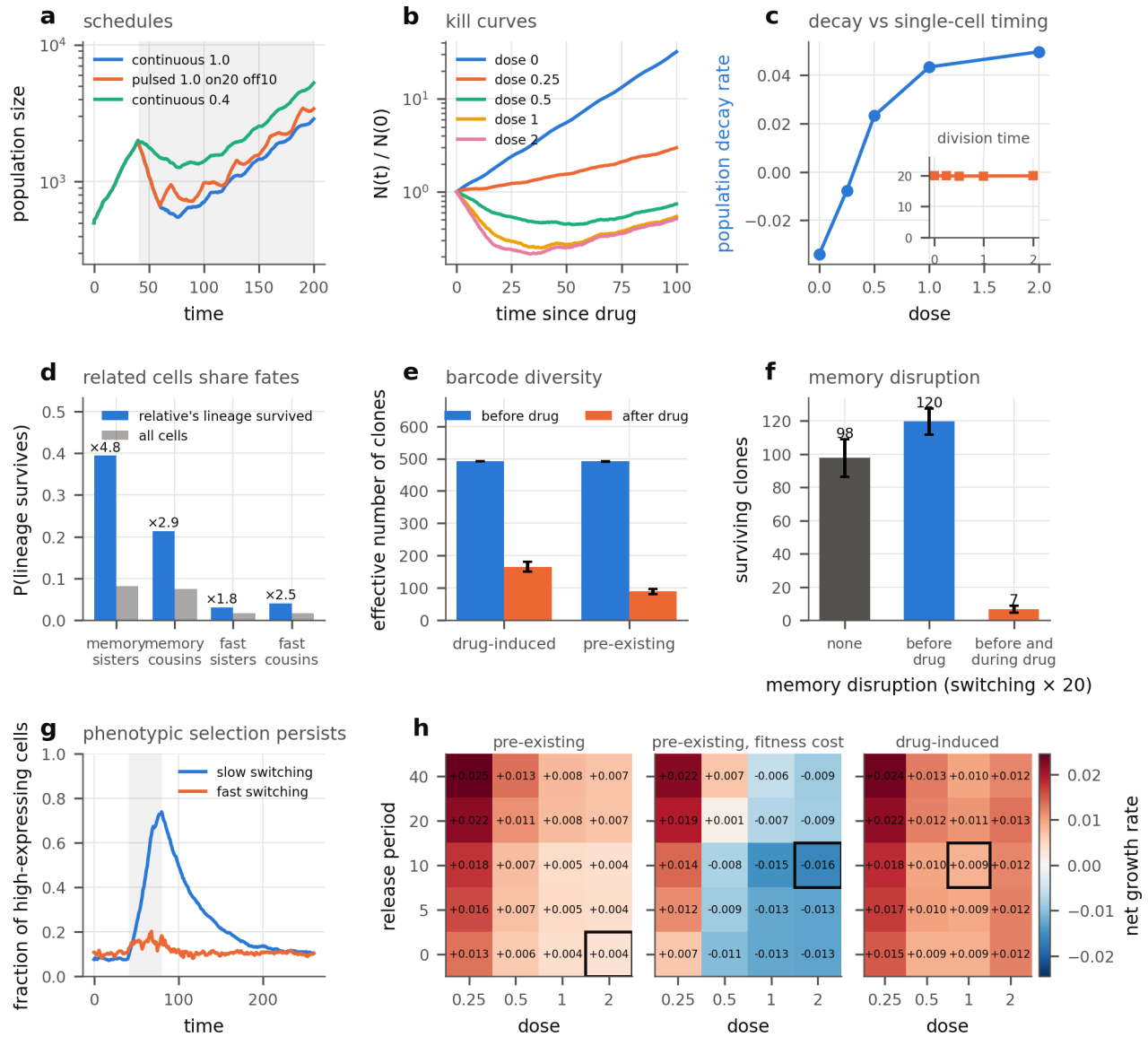


Figure 2. The engine is exact and agrees with an independent implementation. (a) Stationary distribution of a birth-death process (bars: 50 000 simulated cells; line: Poisson). (b) Telegraph model in an intermediate and a bursty regime against the Beta-Poisson law. (c) Bursty protein production against the negative binomial. (d) Kolmogorov-Smirnov distance to the exact law for every kernel (n = 20 000; dotted line: 99% critical value) with wall-clock time per 10⁴ cells. (e) Telegraph model simulated with Biomodelling.jl and with JumpProcesses.jl on the same model; the two-sample KS statistic is below its 99% critical value. (f) Runtime of population simulations against the number of cells (10 genes) and genes (1 000 cells) for the exact and hybrid kernels.



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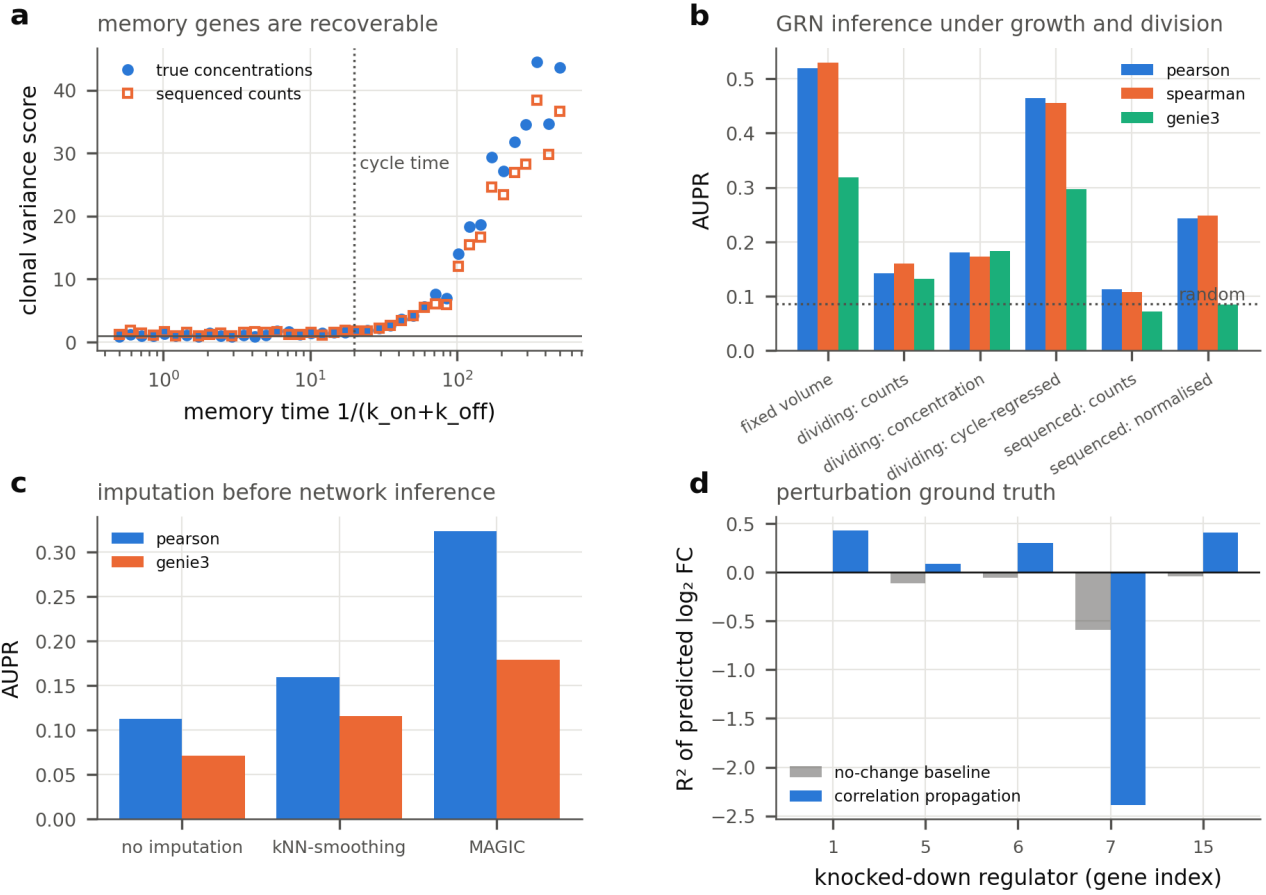
463 **Figure 3. Growth, division and partitioning generate memory, size scaling and cell-cycle effects.** (a)
 464 A single lineage: volume, mRNA and protein of a slowly switching gene, with the promoter-on state shaded. (b)
 465 mRNA count against cell volume for a fast-switching gene without replication (points, line: mean concentration
 466 $\times V$) and with gene replication at mid-cycle (binned means). (c) Correlation of protein concentration at division
 467 between mothers and daughters, sisters and cousins as a function of the promoter switching rate; dotted: one over
 468 the cell-cycle time. (d) Memory timescale in time units against the measured cell-cycle time for two switching
 469 rates; dotted lines: $1/(k_{\text{on}} + k_{\text{off}})$. (e) Coefficient of variation squared of a stable protein's concentration
 470 across the population and along single lineages, for increasing partitioning noise and cell-cycle-time noise. (f)
 471 Telegraph parameters fitted to cells before replication (one gene copy), after replication (two copies) and pooled;
 472 dotted: per-allele values.



473

474 **Figure 4. A single resistance gene with slow promoter switching reproduces persister biology.** (a)
 475 Population size under continuous, pulsed and reduced-dose schedules (shading: drug present). (b) Kill curves at
 476 increasing dose. (c) Net population decay rate over the first 30 time units of treatment against dose, with the
 477 mean division time of cells born under drug (inset), which is unchanged. (d) Probability that a cell's lineage
 478 survives the treatment (at least one descendant alive at the end of the run) given that its sister's or cousin's
 479 lineage survived, against the marginal survival probability, for cells born in the 25 time units before the drug;
 480 memory gene and fast-switching control, with the fold enrichment above each bar. (e) Effective number of clones
 481 (barcodes) before and after drug for pre-existing and drug-induced tolerance (mean $\pm$ s.d. of three seeds). (f)
 482 Surviving clones after drug without memory disruption, with promoter switching accelerated 20-fold from two
 483 cell-cycle times before the drug until the drug arrives, and with the acceleration maintained during exposure
 484 (mean $\pm$ s.d. of five seeds). (g) Fraction of high-expressing cells before, during and after a drug pulse for slow
 485 and fast switching (MGMT-like phenotypic selection). (h) Net population growth rate over a 200-time-unit course
 486 as a function of release period (time units without drug between 20-time-unit exposures) and dose, for pre-existing
 487 tolerance without and with a fitness cost of the resistant state and for drug-induced tolerance; the box marks the

488 most effective schedule.



489

490 **Figure 5. Ground-truth benchmarks.** (a) Clonal variance score against the known memory time of 40
 491 independent genes, from true concentrations and from sequenced counts; dotted: cell-cycle time. (b) Area under
 492 the precision-recall curve of correlation (Pearson, Spearman) and GENIE3 network inference on the same 30-gene
 493 network observed at fixed volume, in a dividing population (raw counts, concentrations, and concentrations
 494 regressed on the cell-cycle covariates copy number and age) and after sequencing (raw and library-size normalised
 495 counts); dotted: random ranking. (c) Effect of kNN-smoothing and MAGIC imputation before inference. (d)
 496 Coefficient of determination of predicted $\log_2$ fold changes after knocking down the five most connected regulators,
 497 for a no-change baseline and for propagation of observational correlations.

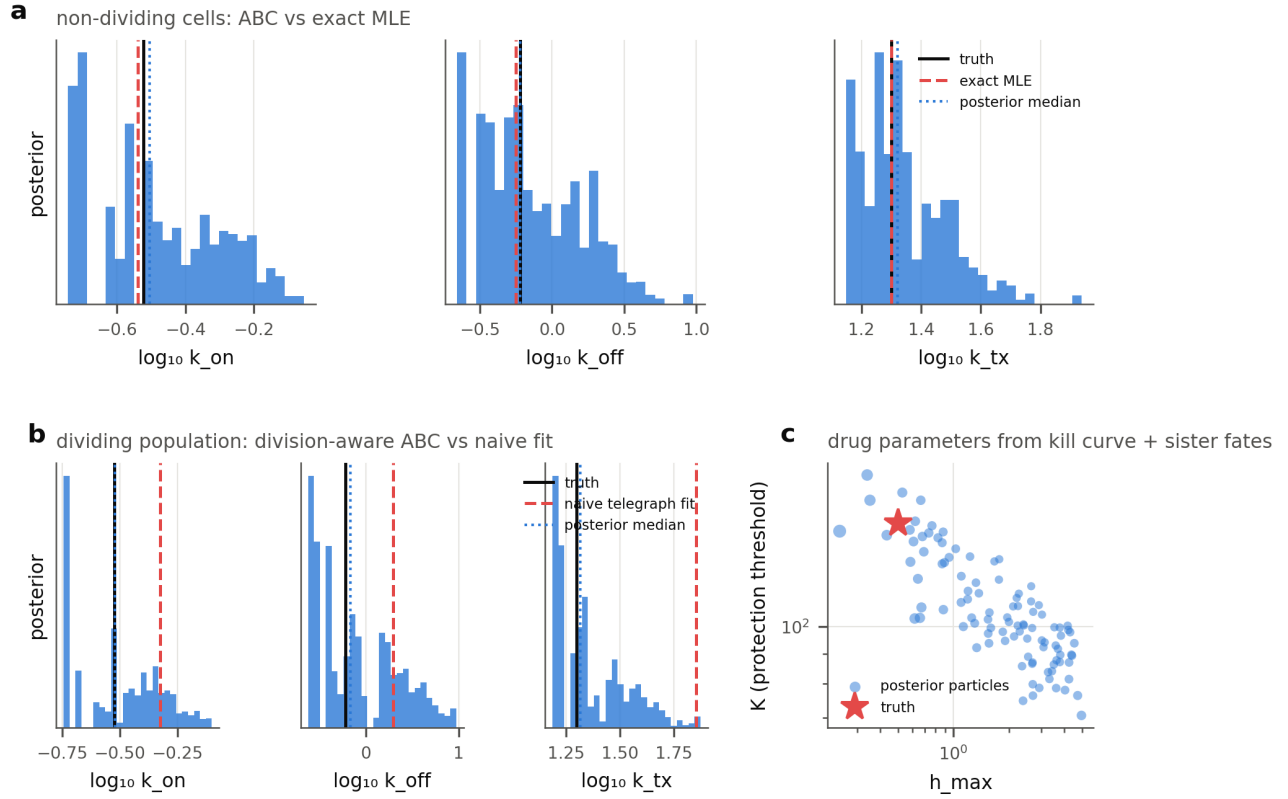
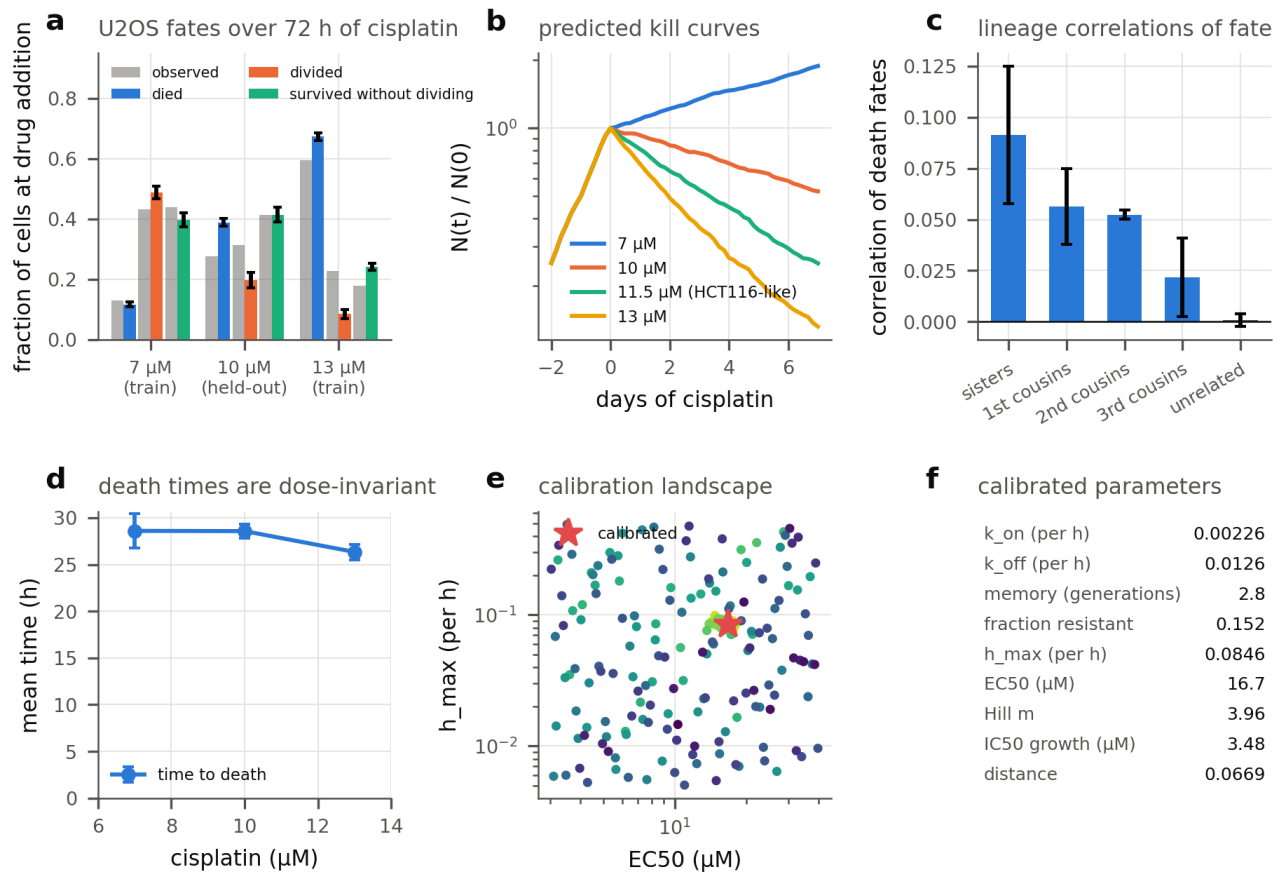


Figure 6. Inference with the forward model. (a) Posterior distributions of telegraph parameters from ABC-SMC on a snapshot of non-dividing cells, with the truth and the exact maximum-likelihood estimate. (b) The same for a snapshot of a growing and dividing population: the naive telegraph fit is biased while division-aware ABC recovers the truth. (c) Posterior particles for the maximal death rate and the protection threshold inferred from a kill curve and sister fate concordance: the two parameters trade off along a ridge that contains the truth. (d) Posterior predictive kill curves simulated at the posterior median against the observed curve.



505

Figure 7. Calibration to laboratory time-lapse data and validation on held-out conditions. (a) Fates over 72 h of cisplatin of U2OS cells present at drug addition, observed by time-lapse microscopy (Iyer et al. 2025) (grey) and simulated by the calibrated model (colours; mean $\pm$ s.d. of four seeds); the model was calibrated on 7 and 13 μM and the 10 μM condition is predicted. (b) Predicted population size at the three concentrations and at 11.5 μM , from two days before the drug to seven days of exposure. (c) Correlation of death fates between sisters, first, second and third cousins present at drug addition, and between unrelated cells (mean $\pm$ s.d. of three seeds), against the observation that correlations extend to second but not third cousins. (d) Mean time to death of cells present at drug addition, and mean division time where at least 20 cells born under drug divided, against concentration (mean $\pm$ s.d. of four seeds). (e) Calibration landscape: every parameter set evaluated by the optimiser, coloured by its distance to the training data. (f) Calibrated parameters.

Figure 8. Dose and schedule optimisation against clinical regimens. (a-b) Population size under continuous dosing, the intermittent schedule of the SWOG S1320 trial (8-week lead-in, then 3 weeks off and 5 weeks on) and adaptive therapy (drug withdrawn below 50% of the initial size, resumed at the initial size) for a melanoma-like population with a rare, heritable pre-resistant state, without (a) and with (b) a fitness cost of the resistant state. (c) Time to progression (20% regrowth from the nadir; mean $\pm$ s.d. of four seeds) for the three schedules and two mechanisms. (d) Time to progression over intermittent schedules of given period and duty cycle (every schedule evaluated by the optimiser; star: best), for the two mechanisms. (e) Population size under the standard (5 of 28 days) and dose-dense (21 of 28 days) temozolomide regimens of the RTOG 0525 trial and a 21-day regimen at the standard cumulative dose, simulated with one-compartment pharmacokinetics (elimination half-life 2.1 h) for MGMT-methylated (1% MGMT-expressing cells) and unmethylated (30%) populations. (f) Depth of response over six cycles (mean $\pm$ s.d. of three seeds). (g) Final population size against the number of

dosing days per 28-day cycle at equal cumulative dose.

Data and code availability

Biomodelling.jl 2.0 is available under the MIT license at <https://github.com/ayoubblasri/Biomodelling.jl>. The laboratory and clinical reference values used for calibration are transcribed, with provenance, in the `paper/data/` directory of the repository. All other data in this article are synthetic and are regenerated, with fixed seeds, by the scripts in the `paper/` directory of the repository (`julia paper/run_all.jl` followed by `python paper/plot_all.py`); the generated tables are archived with the release.

Author contributions

A.L. and M.S. conceived the project. [To be completed.]

Competing interests

The authors declare no competing interests.

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